A gap is not a graviton: strong-coupling exclusions for lattice gravity in first-order form

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In Yang–Mills theory a mass gap is the goal; in gravity it is a diagnosis. We prove, in Lean 4 with Mathlib, that a family of reflection-positive lattice theories whose transfer-matrix gap is bounded below in physical units along a continuum tuning has no continuum limit with a massless mode, and that a massless mode makes the two-point function decay more slowly than any exponential. Combined with the strong-coupling theorems of lattice Yang–Mills for every compact simple group, this excludes the strong-coupling phase of Euclidean lattice gravity written as a gauge theory of the Lorentz group $SO(4)$ (for $\beta_{\mathrm{W}} < 4/29$) or of the de Sitter group $SO(5)$ (for $\beta_{\mathrm{W}} < 15/58$, without compensator) from being gravitational: these phases have a unique state, a positive transfer matrix (so no ghost states), and a gap. We also prove in Lean that an antiscreening gravitational correction to the Yang–Mills β-function preserves asymptotic freedom while a screening one produces an interacting ultraviolet fixed point; that on the 600-cell a gauge-fixing forest can lower the single-link Dobrushin partner count only from 10 to 9, so $\beta_{\mathrm{W}} < 4/9$ is optimal there; and that the MacDowell–Mansouri plaquette-pair action leaves 30 partners per link in the same gauge. We determine exactly the spectrum of the two-dimensional causal-triangulation transfer matrix: $\lambda_n = x_-^{2n+2}/g$, geometric with ratio $(1-s)/(1+s)$, $s = \sqrt{1-4g^2}$, so the lattice gap is of order the spacing, as a continuum gap requires. Two-dimensional Yang–Mills on a Brownian disk has the boundary Wilson loop $d_F(1+gL\sqrt{C_F})e^{-gL\sqrt{C_F}}$ in the quantum length L. On small SU(2) plaquette clusters read as spin networks, every harmonic gap is $\sqrt{2\mu_{\min}}$ for the plaquette Laplacian $\partial\partial^T$, which proves the N-plaquette strip formula; the largest eigenvalue of the triple-product volume operator approaches the classical regular tetrahedron as $1 - 3/(4j)$; and the Barbero–Immirzi parameter is certified to 25 digits for the three standard puncture countings. Nothing here identifies a gravitational phase or constructs a continuum theory.

I. INTRODUCTION

Four-dimensional lattice Yang–Mills theory has a unique state and a transfer-matrix gap at strong coupling for every compact simple gauge group [1], and the gap problem in physical units is reduced there to a single cube estimate. Gravity is the next force to add [2]. The present note starts from one observation: the graviton is massless, so in any approach where gravity is a lattice theory with a transfer matrix, a gap in physical units is a proof that the phase is *not* gravitational. The strong-coupling machinery of [1] produces such gaps at explicit couplings, and some lattice formulations of gravity are gauge theories of compact groups to which it applies as stated.

The formulations are first-order (Palatini) gravity, whose connection is an $SO(4)$ gauge field in Euclidean signature, and MacDowell–Mansouri gravity [3], an $SO(5)$ gauge field broken to $SO(4)$ by a compensating vector [4]. Monte Carlo simulations of such models find a vanishing tetrad at strong coupling [5], and causal and Euclidean dynamical triangulations have non-geometric phases [6]; none of this is proved. The pure Lorentz sector of a Wilson plaquette action generates $R_{\mu\nu}R^{\mu\nu}$ [7, 8], the curvature-squared sector of quadratic gravity, whose spin-2 ghost is conjectured to be confined at strong coupling [9, 10].

All Lean statements are in the repository `romanov360/quantum-gravity`, directory `QuantumGravity/`, built against Mathlib (Lean v4.33.0), without `sorry`, with the standard axioms only; numerical checks are in `python/qg/`.

II. A PHYSICAL GAP EXCLUDES A MASSLESS MODE

Let ρ_a be the finite spectral measure, in lattice units, of the transfer matrix at spacing a for a spatially smeared gauge-invariant field, so that the temporal two-point function is $S_a(t) = \int e^{-Et/a}\, d\rho_a(E)$. A gap m in physical units along the tuning means $\rho_a((-\infty, ma)) = 0$.

Theorem 1 (`physicalGap_ratio_le`, `limit_ratio_of_physicalGap`)**.** *For $0 \le t_1 \le t_2$, $S_a(t_2) \le e^{-m(t_2-t_1)} S_a(t_1)$ for every a, and if $Z_a S_a(t_i) \to L_i$ along any filter for positive renormalisations Z_a, then $L_2 \le e^{-m(t_2-t_1)} L_1$.*

A massless particle in a reflection-positive theory gives the smeared field spectral weight at arbitrarily small energies: a packet contains momenta $|p| < \varepsilon$ and $E = |p|$. We call a finite measure σ on $[0,\infty)$ *massless* if $\sigma([0,\varepsilon)) > 0$ for every $\varepsilon > 0$.

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Theorem 2 (`massless_not_expDecay`). *If σ is massless, then for no $\Delta > 0$ and C does $\int e^{-Et} d\sigma(E) \le Ce^{-\Delta t}$ hold for all $t \ge 1$.*

Idea. $\int e^{-Et} d\sigma \ge \sigma([0, \Delta/2))\, e^{-\Delta t/2}$, and $Ce^{-\Delta t/2} \to 0$. □

Theorem 3 (`no_massless_limit_of_physicalGap`). *If a family of lattice theories has a gap $m > 0$ in physical units along the tuning and its renormalised two-point functions converge at every $t > 0$ to $\int e^{-Et} d\sigma(E)$ for a finite measure σ on $[0, \infty)$, then σ is not massless.*

The other alternative, a gap bounded below in *lattice* units, is the no-go theorem of [1]: every renormalised two-point function then vanishes in the limit. A graviton therefore needs $\Delta_{\mathrm{lat}}(a)/a \to 0$.

III. FIRST-ORDER LATTICE GRAVITY AT STRONG COUPLING

The theorem of [1] on every compact simple group gives, in the forest gauge, a unique free-boundary state, exponential clustering and a transfer-matrix gap for $\beta_{\mathrm{W}} = N^2\beta < 2N\,\mathrm{Ric}/29$, where N is the dimension of the representation in the Wilson action and $\mathrm{Ric} = c/4$ when the Killing form is c times the trace form. The constants are quadratic-form identities on a Cartan subalgebra, proved for all Cartan elements in `GravityGroups.lean`:

group (rep.)	role	c	Ric	range of β_{W}
$SU(2)$ (2)	self-dual Spin(4)	4	1	4/29
$SO(4)$ (4)	Lorentz, tetrad index	2	1/2	4/29
$SO(5)$ (5)	de Sitter (MM)	3	3/4	15/58
$Sp(2)$ (4)	de Sitter, spinor	6	3/2	12/29

$SO(4)$ is semisimple; its two factors have the same Ricci constant, which is all the Bakry–Émery step uses.

Corollary 4. *The Euclidean lattice Lorentz gauge theory with weight $\exp(4\beta \sum_p \mathrm{Tr}_4 U_p)$, $U \in SO(4)$, has for $\beta_{\mathrm{W}} = 16\beta < 4/29$ a unique infinite-volume state, exponential clustering and a transfer-matrix gap. Its transfer matrix is a positive operator (reflection positivity of the Wilson action), so its Hilbert space has no negative-norm states. If the gap persists in physical units along a tuning, the limit has no massless mode in any channel (Theorem 3).*

The centre element $-1 \in SO(4)$ acts as -1 on the vector representation, so vector Wilson loops have an area law at strong coupling [11]. The fundamental character coefficient (`lorentz_strong.py`) is $a_F(\beta_{\mathrm{W}}) = \beta_{\mathrm{W}}/16 + O(\beta_{\mathrm{W}}^3)$ (for $SU(2)$: $I_2(\beta_{\mathrm{W}})/I_1(\beta_{\mathrm{W}}) = \beta_{\mathrm{W}}/4 + \dots$), so the leading-order string tension at $\beta_{\mathrm{W}} = 4/29$ is $-\log a_F = 4.75$ in lattice units.

Remark. In the frozen-tetrad model the curvature-squared sector is therefore ghost-free and gapped at strong coupling; this is the fixed-spacing analogue of ghost confinement in quadratic gravity [9, 10]. Whether a continuum limit of this lattice model is quadratic gravity is not addressed.

For MacDowell–Mansouri gravity [3] the action pairs each plaquette with the complementary plaquette at the same site. In the forest gauge of [1] every free link then has exactly 30 partners with multiplicity (42 without gauge fixing; `MMPartners.lean`, by `decide`), against 12 for the Wilson action. The Lipschitz constant of a pair term and the compensator remain; the programme is in `docs/G2-pregeometric-phase.md`.

IV. GRAVITY AND ASYMPTOTIC FREEDOM

Above the Planck scale the Yang–Mills coupling runs as $dg/dt = -bg^3 - fg$, with f a gravitational coefficient whose sign is scheme dependent but non-negative in symmetry-preserving computations [12]. In $u = 1/g^2$, $u(t) = (u_0 + b/f)e^{2ft} - b/f$.

Theorem 5 (`RunningWithGravity.lean`). *(i) u solves $du/dt = 2b + 2fu$. (ii) If $f > 0$, $b \ge 0$, $u_0 \ge 0$ then $u(t) \ge u_0 + 2bt$: the coupling is never larger than without gravity, and $g^2 \to 0$. (iii) If $f < 0 < b$, then $g^2 \to -f/b > 0$, an interacting ultraviolet fixed point. (iv) If $f > 0$, every $u_* \ge u_0$ is attained at $t_* = \log \frac{u_* + b/f}{u_0 + b/f}/(2f) \ge 0$.*

Item (iv) is the statement of [1] that the intermediate coupling window is entered at a spacing-independent scale, with the gravitational correction included.

V. YANG–MILLS ON CURVED LATTICES

The single-link Dobrushin count of [1] is local: on any 2-complex with a gauge-fixing forest F, a free link e has $P(e) = \sum_{p \ni e}(\#\text{free links of } p - 1)$ partners and $SU(2)$ is unique for $\beta_{\mathrm{W}} < 4/\max_e P(e)$. The search in `forest_partners.py` reproduces $P = 12$ on $\mathbb{Z}^4$ and $P = 7$ for the period-3 forest on $\mathbb{Z}^3$, and finds by exhaustion that period-2 forests on $\mathbb{Z}^3$ cannot go below 8.

Proposition 6 (`cell600_nine_optimal`). *On the 600-cell (120 vertices, 720 edges, every edge in 5 triangles) every forest leaves some free link with at least 9 partners, and a forest with $\max P = 9$ exists. The optimal single-link $SU(2)$ range is $\beta_{\mathrm{W}} < 4/9$.*

Proof. $P(e) = 10 - n(e)$, where $n(e)$ is the number of fixed links sharing a triangle with e. A forest has at most 119 links, each a neighbour of 10 links, so $\sum_{e \notin F} n(e) \le 1190 < 2 \cdot 601$; some free link has $n(e) \le 1$. A greedy forest attains 9. □

For closed triangulations a forest has $V - 1$ links among $E \approx 6V$, so gauge fixing helps little and the range is governed by the maximal edge degree; a uniform statement over dynamical triangulations needs a degree cap.

VI. TWO EXACTLY SOLVABLE MODELS

a. YM_2 on a Brownian disk. Two-dimensional Yang–Mills depends on the surface only through its area [13, 14]; the free Brownian disk of perimeter L has area density $L^3(2\pi A^5)^{-1/2}e^{-L^2/2A}$ [15]. Hence $\mathbb{E}[e^{-sA}] = (1+L\sqrt{2s})e^{-L\sqrt{2s}}$ (an inverse-Gamma Laplace transform via $K_{3/2}$), and the Haar-averaged boundary Wilson loop $d_F e^{-g^2 A C_F/2}$ becomes

$$\langle\!\langle W_F \rangle\!\rangle_L = d_F\,(1 + gL\sqrt{C_F})\,e^{-gL\sqrt{C_F}},$$

a perimeter law in quantum length with tension $\sqrt{2\sigma_F}$. Since $\int Z(U,A)\,dU = 1$ for every A, quenched and annealed averages agree. On the sphere each representation shifts the cosmological constant by $g^2 C_\lambda/2$, and only the trivial one is non-analytic at $\mu = 0$: the string susceptibility exponent $-1/2$ is unchanged. Monte Carlo checks are in `ym2_brownian.py`.

b. 2D causal triangulations. The one-step transfer matrix of 2D CDT is $M(l_1,l_2) = g^{l_1+l_2-1}\binom{l_1+l_2-1}{l_1-1}$ [16], $0 < g < \frac{1}{2}$.

Theorem 7 (`CDT2D.lean`; proof in `docs/G3-cdt.md`). *Let $s = \sqrt{1-4g^2}$ and $x_- = (1-s)/(2g)$. The symmetrised transfer matrix $D^{-1/2}MD^{1/2}$, $D = \mathrm{diag}(l)$, is a positive Hilbert–Schmidt operator on ℓ^2 whose nonzero spectrum is $\lambda_n = x_-^{2n+2}/g$, $n \geq 0$. Hence $\lambda_{n+1}/\lambda_n = (1-s)/(1+s)$ and $\Delta_{\mathrm{lat}} = 2\,\mathrm{artanh}\,s$; with $g = \frac{1}{2}e^{-a^2\Lambda}$, $\Delta_{\mathrm{lat}}(a)/a \to 2\sqrt{2\Lambda}$.*

Idea. On $v_x = (x^l)_l$, $v_x M = h(x)\,v_{\varphi(x)}$ with $h = x/(1-gx)$ and the Möbius map $\varphi = g/(1-gx)$ (proved in Lean from the negative binomial series). Differentiating n times at the fixed point $x_- \in (0,1)$, where $\varphi'(x_-) = x_-^2$ and $h(x_-) = x_-^2/g$, makes M triangular on $w_k = \partial^k v_x|_{x_-}$ with diagonal x_-^{2n+2}/g. The $w_k D^{1/2}$ lie in ℓ^2 and span a dense set (an analytic function on the unit disk with all derivatives zero at x_- vanishes), and a compact self-adjoint operator is diagonalised by them. □

This is a solvable model of quantum gravity in which the lattice gap scales with the spacing, as [1] requires of a theory with a continuum gap; 2D gravity has no graviton, so Theorem 3 is silent.

VII. SPIN NETWORKS ON SMALL CLUSTERS

Hartshorn [17] reads the exactly solved SU(2) Hamiltonian clusters as loop-quantum-gravity kinematics: the link Casimir $j(j+1)$ is a squared area in units of $8\pi\gamma\ell_P^2$, and the gauge-invariant states are spin networks. Three of the open items listed there have the following answers.

a. Harmonic gaps of any cluster. Abelianising at weak coupling, plaquette fluxes $B = \partial A$ are oscillators with kinetic matrix $K = \partial\partial^T$ on plaquettes, and $2 - W_p \simeq |B_p|^2/4$ gives $\omega^2 = \mu/2$, $\mu \in \mathrm{spec}\,K$ (threefold). The lowest gauge-invariant level is two soft quanta: $\Delta_{g\to 0} = \sqrt{2\mu_{\min}}$, with $\mu_{\min}$ the least nonzero eigenvalue (zero modes are Bianchi identities of closed surfaces). For planar clusters $K = 4I - A$ with A the adjacency of plaquettes sharing a link. This reproduces $\sqrt{8}$, $\sqrt{6}$ and $2\sqrt{2-\cos\frac{\pi}{4}}$ for one, two and three plaquettes, explains why the L-shape has the strip's gap (both graphs are P_3), and predicts $\Delta = 2$ for the 2×2 block and $2\sqrt{2-\sqrt{3}/2}$ for the T-shape. For the N-strip the eigenvectors are sines, and `ClusterModes.lean` proves $\Delta_{g\to 0} = 2\sqrt{2-\cos\frac{\pi}{N+1}}$ for every N.

b. The volume of a four-valent vertex. In the coupled basis $|k\rangle$ of $(j_1 j_2) \to k$, the triple product $Q = \epsilon_{abc}J_{1a}J_{2b}J_{3c}$ is i times a tridiagonal matrix with $a_k = \sqrt{\Delta(j_1,j_2,k)\Delta(j_3,j_4,k)}/(4\sqrt{4k^2-1})$, $\Delta(a,b,k) = (a+b+k+1)(-a+b+k)(a-b+k)(a+b-k+1)$, which agrees with brute-force tensor products to 10^{-14}. Matching $V^2 = \frac{2}{9}|A_1\cdot(A_2\times A_3)|$ of a classical tetrahedron fixes $V = \frac{\sqrt{2}}{3}(8\pi\gamma\ell_P^2)^{3/2}\sqrt{|Q|}$. For four equal spins the ratio of $\lambda_{\max}(Q)$ to the regular tetrahedron's $\frac{4}{3\sqrt{3}}[j(j+1)]^{3/2}$, which [17] found rising to 0.921 at $j = 8$, tends to 1: numerically $j(1-\mathrm{ratio}) \to 3/4$ (0.74998 at $j = 65536$), a semiclassical limit of the kind derived by Bohr–Sommerfeld methods [18].

c. The Immirzi parameter. Matching $S = A/4\ell_P^2$ to puncture counts gives $\gamma = \ln 2/(\pi\sqrt{3})$ for spin-$\frac{1}{2}$ punctures, and for all spins the roots of $1 = \sum_j c_j e^{-2\pi\gamma\sqrt{j(j+1)}}$ with $c_j = 2$ (Domagala–Lewandowski, Meissner [19, 20]) or $c_j = 2j+1$ (Ghosh–Mitra [21]). Bisection with an explicit geometric tail bound certifies $\gamma = 0.237532957965920664358377 1$ and $0.27406685823282999621676 37$.

VIII. COSMOLOGICAL BILLIARDS

Near a spacelike singularity, vacuum gravity becomes a billiard for the hyperbolic Kac–Moody algebra AE_3, and eleven-dimensional supergravity one for E_{10} [22]. `Billiards.lean` checks by `decide` the leading principal minors $2, 3, -2$ of AE_3 and $2, 3, \ldots, 10, -1$ of E_{10} (one sign change: Lorentzian signature), and `bkl_billiard.py` that the BKL era lengths are the continued-fraction digits of the initial u, with Gauss–Kuzmin statistics.

IX. WHAT IS NOT CLAIMED

No result here identifies a gravitational phase, constructs a continuum limit, or concerns Lorentzian signature. The strong-coupling statements are at fixed lattice spacing; Theorem 3 needs the gap along a tuning, which is not proved for any of the models above. The open problems are listed in `OPEN.md`.

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